Sequential Conditioning Beats Parallel Sampling: A Matched-Compute Decomposition of Debate on Small Qwen Models

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Abstract

Multi-agent debate is often presented as evidence that multiple model instances can reason better together than one model can alone. Recent matched-compute studies challenge that interpretation, showing that independent sampling and self-consistency often close the gap. We revisit the question on small Qwen-family reasoning models and find a different mechanism: the useful ingredient is not usually the agent boundary, but the fact that later reasoning is conditioned on earlier reasoning. On Qwen3-1.7B and Qwen3-4B, sequential scaffolds such as debate, Self-Refine, and long chain-of-thought beat $k{=}4$ self-consistency by +10 to +24 points on MATH-500 and +18 to +23 points on GPQA-Diamond. Debate also exceeds the selector over four independent samples, so the gain cannot be explained as better aggregation of those samples. Decomposing the scaffold shows that role asymmetry is unnecessary on math, where Self-Refine and long chain-of-thought match debate within statistical noise, but can help on science multiple-choice at base checkpoints. A negative Llama-3.1-8B probe bounds the claim. The result is a scoped counterexample to the matched-compute debate critique, and a simpler lesson: for these models, conditioning the next attempt on the previous one can matter more than drawing more independent attempts.

1 Introduction

Multi-agent debate is attractive because it turns a single model's answer into a conversation. A first model proposes a solution, a second model critiques it, and a later turn can repair the error. The trouble is that a conversation is not free. If the same generation budget were spent on several independent attempts, self-consistency might recover the answer without any agent structure at all. This matched-compute objection has become the central critique of debate-style inference methods [1–3].

This paper asks what remains after taking that critique seriously. We study small Qwen-family thinking-mode-trained models on math and science reasoning, and compare two ways of spending the same budget. The first is *parallel*: draw four independent solutions and aggregate them. The second is *sequential*: let later reasoning condition on earlier reasoning, either through debate, self-critique, or one long chain-of-thought.

The empirical pattern is simple. On Qwen3-1.7B and Qwen3-4B, sequential scaffolds beat matched-compute self-consistency by +10 to +24 points on MATH-500 and +18 to +23 points on GPQA-Diamond. More importantly, debate exceeds the pass@4 selector: it solves problems for which all four independent samples are wrong. A judge that reads the same four samples does not reproduce this effect. The gain is therefore not just better selection among parallel answers.

The result is not a blanket endorsement of multi-agent debate. On math, Self-Refine [4] and long chain-of-thought match role-asymmetric debate within statistical noise at base checkpoints. The role

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split matters mainly on GPQA-Diamond, where debate adds 5–8 points over Self-Refine at base checkpoints. A Llama-3.1-8B-Instruct probe reverses the result, with critique scaffolds losing to self-consistency. The claim is therefore scoped: this is a Qwen-family small-model phenomenon, and the central mechanism is sequential conditioning rather than agent multiplicity.

We also report a short training-time multi-agent probe that did not outperform single-agent GRPO and showed clear Verifier pathologies. That negative result is included for completeness, but the main contribution is the inference-time decomposition: conditioning the next attempt on the previous one can move the model outside the support of a small independent sample set, while explicit agent roles add value only conditionally.

2 Related work

Multi-agent debate as inference scaffolding. Multi-agent debate and related agent systems have been used as inference-time scaffolds and as training-time objectives [5–11]. The closest methodological critiques argue that many debate gains disappear under matched-compute baselines [2, 1, 3]. We take those critiques as the right standard and identify a narrower regime where the conclusion changes: small Qwen-family models, hard math/science reasoning, and a strict alternating protocol.

Self-Refine and single-agent iterative critique. Self-Refine [4] showed that a model can improve by critiquing and revising its own outputs. We treat Self-Refine as a central baseline rather than as related color: if Self-Refine matches debate, then the role split is not the main source of the gain. Our additions are the matched self-consistency baseline, pass@4 selector analysis, paired bootstrap intervals, and cross-checkpoint/cross-domain replication. Reflexion [12] and CRITIC [13] make adjacent iterative-critique claims; we cite them but do not run them as full arms.

Multi-agent debate distillation. IMAD [14], AgentArk [15], and Chain-of-Agents [16] distill debate trajectories into a single model via post-debate SFT or RL. Our setting is orthogonal: we measure the inference-time-only marginal value of scaffolding on policies that have not been distilled. Our findings suggest the distillation effort may be lower-value than the literature assumes, since the un-distilled base already supports a strong scaffold at inference.

Training-time multi-agent RL. MARTI, Stronger-MAS, AT-GRPO, MAPoRL, CRM, and MAGRPO [5, 7, 6, 8, 17] train multiple agents with joint or per-agent rewards. Our short training-time probe belongs to this lineage, but it is not the main claim of the paper; details are deferred to Appendix E.

Self-consistency and parallel sampling. Self-consistency [18] samples k independent responses and majority-votes. Pass@k is the corresponding oracle selector: it asks whether any of those independent samples contains the right answer. This paper's central empirical contrast is between those parallel samples and methods that condition later reasoning on earlier reasoning.

Outcome-RL limits. [19] argues that outcome-RL improves pass@1 without necessarily expanding the underlying reasoning capacity of the base model. Our base and GRPO-trained Qwen3-4B checkpoints converge to the same debate accuracy on MATH-500, which is consistent with that picture.

3 Method

3.1 Trained Solvers

We compare two trained policies on the same architecture (Qwen3-4B):

- **Base.** The untrained `Qwen/Qwen3-4B` instruct checkpoint, and `Qwen/Qwen3-1.7B` for the cross-scale comparison.
- **Solo-GRPO.** Qwen3-4B trained with outcome-only single-agent GRPO on `hendrycks/competition_math` TRAIN L3–5 (disjoint from the MATH-500 test set) for 30 PPO updates: $K = 8$ rollouts/problem, response budget 2048, KL-anchored at $\lambda_{KL} = 0.001$ with `low_var_kl` [20], repetition penalty 1.15.

3.2 Inference matrix

Given a trained Solver checkpoint θ, we evaluate three families of inference procedure (Table 1). The single-shot baseline measures the model's ordinary pass@1 behavior. The parallel baselines spend the extra budget on independent samples, either by majority vote, by the any-correct pass@4 selector, or by asking a judge to read all four samples. The sequential scaffolds spend the budget by conditioning later reasoning on earlier reasoning. All procedures use Solver temperature 0.7, Verifier temperature 0.3 in debate variants, `repetition_penalty=1.15`, and the same sympy-aware answer matcher. Multi-turn procedures cap each of four turns at 2048 tokens; self-consistency and pass@4 draw four 2048-token samples; long chain-of-thought uses one 8192-token cap. The judge-over-samples baseline uses the same checkpoint as the Solver: it first generates the same four samples as self-consistency, then gives their full rationales to a judge prompt that asks the model to select or reconstruct the correct answer and end with a boxed response.

Table 1: Inference procedures. "Budget" is the matched per-problem generation cap in tokens.

Family	Procedure	Structure	Budget
Baseline	single-shot	one Solver sample	2048
Parallel	self-consistency	four Solver samples, majority vote	8192
Parallel	pass@4 selector	four Solver samples, any-correct	8192
Parallel	judge-over-samples	four samples + one judge turn	10240
Sequential	long chain-of-thought	one long Solver sample	8192
Sequential	Self-Refine [4]	generate, critique, revise, critique	8192
Sequential	role-asymmetric debate	Solver and Verifier alternate	8192
Sequential	frozen-critic debate	Solver with frozen base Verifier	8192
Sequential	symmetric peer review	two Solver-prompted agents alternate	8192

Realized output tokens differ across arms despite a matched cap. Measured on solo-GRPO MATH-500, debate uses about 3.6k tokens and Self-Refine about 4.0k, while self-consistency uses about 6.7k because each of the four samples is a fresh chain-of-thought. The sequential scaffolds therefore do not win by consuming more realized output.

4 Experimental setup

Models. Three Qwen-family checkpoints: Qwen3-1.7B base (cross-scale), Qwen3-4B base (primary), Qwen3-4B + solo-GRPO step-30. A short training-time multi-agent probe is summarized in Section 6 and detailed in Appendix E. The cross-family probe in Appendix B uses Llama-3.1-8B-Instruct.

Eval datasets. MATH-500 (full, $n = 500$, stratified L1–L5), AIME 2024 ($n = 30$), AIME 2025 ($n = 30$), GPQA-Diamond ($n = 198$, Diamond split).

Eval pipeline. Standalone vLLM 0.8.4, V1 engine, bf16, one H200. Sympy-aware answer matching; comparator patch (Section 3.2) rescores all transcripts to handle unit-suffix mismatches like 30° vs 30. Per-problem JSONL records, prompts, and reproduction commands are preserved in the repository.

Bootstrap intervals. We report per-cell 95% percentile bootstrap intervals for paired differences between arms, using 2000 resamples per cell. For aggregate cells with $n = 500$, we also cross-check against the standard Wald interval on the difference of proportions.

5 Results

5.1 Sequential scaffolds beat matched-compute self-consistency

Table 2 gives the headline comparison: single-shot, self-consistency, the pass@4 selector, and four-turn debate. Full per-level results are in Appendix C.

Debate beats self-consistency on every checkpoint and on the hardest cells within each checkpoint. The largest margin is on base Qwen3-4B MATH-500 L5, where debate gains 41.8 points. The two base models have similar MATH-500 margins despite a 2.4× parameter gap, suggesting that the effect is not simply a scale artifact within this small-model range.

Table 2: Headline cells, post-rescore. "Self-cons." is self-consistency at $k = 4$; "pass@4" is the any-correct selector over the same four samples. Δ is debate minus self-consistency. Bootstrap intervals are paired, 2000 resamples.

Cell	Single	Self-cons.	pass@4	Debate	Δ	95% interval
Base Qwen3-4B						
MATH-500 overall	0.544	0.622	0.706	**0.858**	+23.6	[+19.4, +27.8]
MATH-500 L5	0.299	0.358	0.478	**0.776**	+41.8	[+33.6, +50.7]
AIME 2024	0.033	0.100	0.133	**0.367**	+26.7	[+13.3, +43.3]
AIME 2025	0.067	0.000	0.133	**0.333**	+33.3	[+16.7, +50.0]
Solo-GRPO step-30						
MATH-500 overall	0.668	0.752	0.798	**0.858**	+10.6	[+7.4, +14.0]
MATH-500 L5	0.507	0.575	0.604	**0.784**	+20.9	[+14.2, +28.4]
AIME 2025	0.100	0.133	0.133	**0.367**	+23.3	[+10.0, +40.0]
Base Qwen3-1.7B						
MATH-500 overall	0.528	0.572	0.688	**0.796**	+22.4	[+18.4, +26.4]
MATH-500 L5	0.261	0.299	0.440	**0.634**	+33.6	[+25.4, +41.8]
AIME 2025	0.000	0.067	0.100	**0.267**	+20.0	[+6.7, +33.3]

The base Qwen3-4B and the solo-GRPO step-30 checkpoint converge to *identical* accuracy under the debate scaffold (0.858 vs 0.858 on MATH-500 overall) despite a 12.4-point gap in single-shot accuracy. Outcome RL closes part of the gap to debate, but not all of it: the scaffold extracts a similar terminal ceiling from both. A practitioner without training compute can reach within one point of the GRPO-trained 4B checkpoint's debate accuracy by running the base 4B in the same scaffold.

5.2 Debate exceeds the independent-sample selector

Self-consistency can fail because majority vote chooses the wrong sample. Pass@4 is a stronger baseline: it asks whether any of four independent samples contains the right answer. By construction, it upper-bounds every selector over those four boxed answers. When debate exceeds pass@4, it is not merely choosing better among the boxed answers already produced; it is producing correct answers on problems where the four independent samples are all wrong. Debate exceeds this selector on every below-ceiling cell we test (Table 3). The claim is intentionally narrow: this rules out simple final-answer aggregation over the same four samples, while leaving stronger rationale-level or learned selectors as future baselines.

Table 3: Pass@4 overshoot on representative hard cells, post-rescore. Top block: debate (sequential) vs pass@4. Bottom row, for contrast: judge-over-samples (LLM-aware aggregation of the same four samples at $+25\%$ budget) does *not* exceed pass@4 on math. Full per-cell intervals in Appendix C.

Cell	Arm $-$ pass@4	95% interval
Debate $-$ pass@4		
Base Qwen3-4B MATH-500 L5	**+29.9**	[+21.6, +38.8]
Base Qwen3-4B GPQA-Diamond	+17.2	[+9.6, +24.7]
Solo-GRPO MATH-500 L5	+17.9	[+11.2, +25.4]
Solo-GRPO AIME 2025	+23.3	[+10.0, +40.0]
Qwen3-1.7B MATH-500 L5	+19.4	[+11.2, +27.6]
Qwen3-1.7B GPQA-Diamond	+15.2	[+7.6, +22.7]
Judge-over-samples $-$ pass@4 (contrast)		
Solo-GRPO MATH-500 (overall)	-0.2	$[-3.4, +2.8]$

5.3 What is load-bearing?

The previous sections show that debate beats matched-compute self-consistency and exceeds the selector over four independent samples. The next question is whether the role-asymmetric debate structure is necessary. Table 4 compares the three sequential scaffolds that matter most for this question: one long chain-of-thought, Self-Refine, and role-asymmetric debate.

The decomposition has a clear shape.

Table 4: Sequential scaffold decomposition, post-rescore. Long chain-of-thought has no explicit critique or turn structure; Self-Refine has critique but no role split; debate has both critique and role asymmetry.

Cell	Self-consistency	Long CoT	Self-Refine	Debate
Base 4B MATH-500	0.622	0.856	0.854	0.858
Base 4B GPQA	0.242	0.399	0.414	**0.470**
Qwen3-1.7B MATH-500	0.572	0.798	0.802	0.796
Qwen3-1.7B GPQA	0.157	0.283	0.288	**0.364**
Solo-GRPO MATH-500	0.752	0.818	0.854	0.858
Solo-GRPO GPQA	0.298	0.364	0.460	**0.475**

Self-Refine matches debate on math, but lags on base-checkpoint GPQA. The paired bootstrap interval for Self-Refine minus debate crosses zero on every $n = 500$ MATH-500 cell across all three checkpoints. On GPQA-Diamond, Self-Refine lags debate by 7.6 points on Qwen3-1.7B and by 5.6 points on Qwen3-4B base; on the GRPO-trained checkpoint the gap is not significant. Role asymmetry therefore matters in one clear regime: science multiple-choice at base checkpoints. One possible reason is protocol fit: on Qwen3-1.7B, the Verifier omits the `<verify>` tag on 81% of MATH-500 turns but only about 47% on GPQA. Figure 1 visualizes the Self-Refine comparisons.

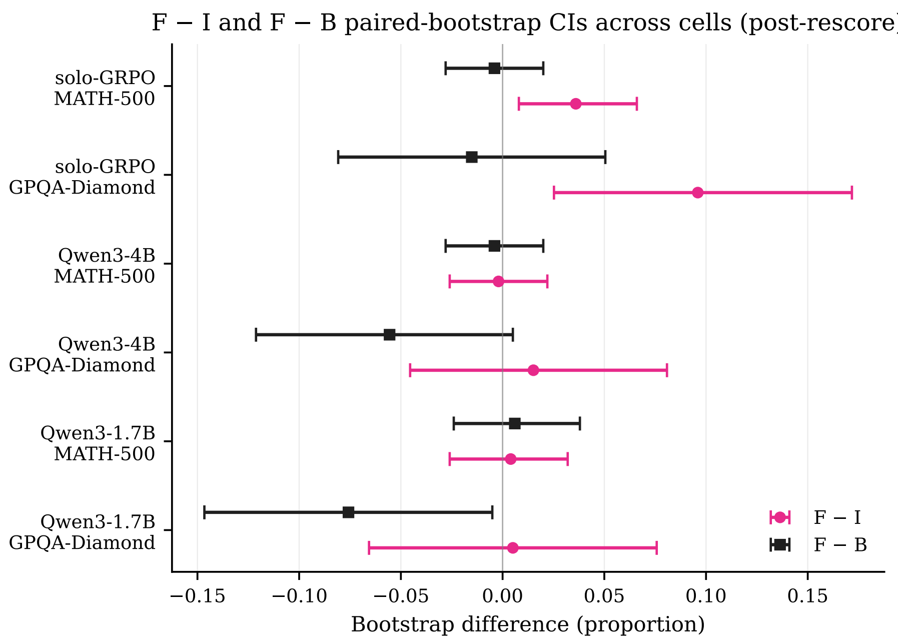

Figure 1: Paired bootstrap 95% intervals for Self-Refine minus long chain-of-thought and Self-Refine minus role-asymmetric debate. Intervals crossing zero indicate no significant difference. Self-Refine matches debate on math, while debate has an edge on base-checkpoint GPQA.

A frozen critic recovers most of the debate gain. In an additional GRPO-only ablation, replacing the Verifier with the frozen Qwen3-4B base still beats self-consistency by 6.2 points overall and 17.1 points on MATH-500 L5. The same-policy debate is 4.4 points higher overall. This suggests that critic quality contributes, but most of the matched-compute gain already comes from sequential critique-then-revise.

Long chain-of-thought recovers most of the math lift. A single 8192-token chain-of-thought, with no turn structure and no explicit critique, reaches 0.818, 0.856, and 0.798 MATH-500 accuracy across the three checkpoints. It is statistically tied with Self-Refine and debate on base-checkpoint math. On GPQA, however, debate beats long chain-of-thought by 7–11 points on every checkpoint.

Sequential conditioning, not aggregation, explains the pattern. Debate, Self-Refine, and long chain-of-thought all exceed pass@4 on the main hard cells. The judge-over-samples arm, despite seeing the same four candidates and using 25% more budget, stays near the pass@4 selector. The simplest mechanism summary is: sequential conditioning, whether implemented as debate, self-critique, or one long reasoning trace, can generate answers outside the four-sample parallel support. Explicit critique structure adds value on top only in domain- and checkpoint-conditional regimes.

A separate per-trajectory analysis (Appendix D) shows the corrective mechanism is real and not noisy: across all three checkpoints debate flips 25–36% of wrong T0 answers to correct, while only 0.2% of correct T0 answers regress. The Verifier's `<verify>` verdict tag is uncorrelated with Solver correctness (on solo-GRPO, the Verifier approves a wrong T0 in 30.8% of cases and rejects it in only 2.2%; the remaining $\sim$67% of wrong-T0 Verifier turns omit the tag entirely). The mechanism is therefore not the binary verdict; it is the Verifier's *reasoning text*, which re-conditions the next Solver turn even when the explicit verdict label is wrong.

5.4 Where self-consistency fails

On base Qwen3-4B AIME 2025, self-consistency at $k = 4$ gets zero of thirty problems correct, worse than the two-of-thirty single-shot baseline. When the model's modal samples are wrong, majority vote selects the dominant wrong answer. Debate lifts the same cells: base AIME 2025 goes from 0.067 single-shot and 0.000 self-consistency to 0.333 debate; solo-GRPO goes from 0.100 single-shot and 0.133 self-consistency to 0.367 debate. Sequential conditioning, including bare long chain-of-thought, does not have the same failure mode because each generation conditions on prior in-context reasoning rather than on independent samples from the same distribution.

5.5 The gain is difficulty-conditioned

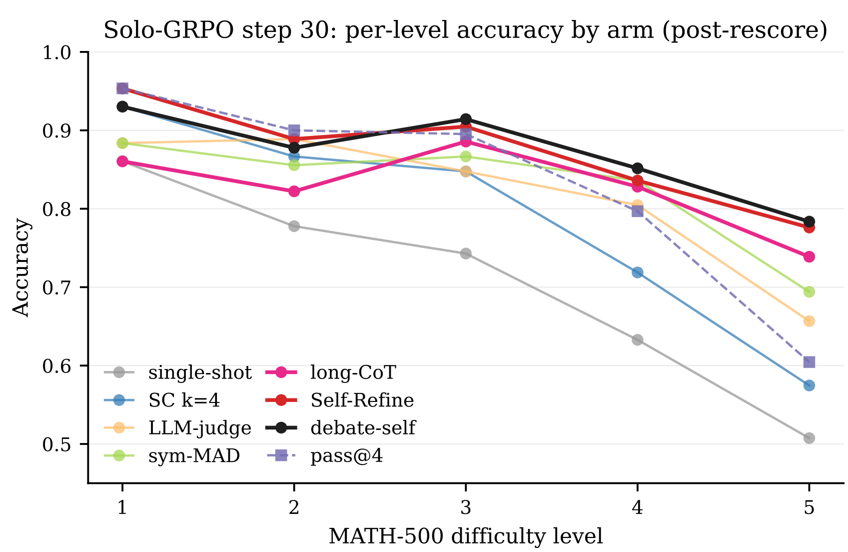

Figure 2: Per-level accuracy on MATH-500 by arm, solo-GRPO step-30 (post-rescore). All arms converge at L1; the sequential scaffolds separate from the parallel-sampling arms at L4–L5, where the model has the most headroom.

The margin grows with difficulty. On solo-GRPO, debate and self-consistency are tied on L1 and nearly tied on L2, but debate is 20.9 points higher on L5. Base Qwen3-4B has more headroom, and the L5 margin reaches 41.8 points. The same pattern appears on Qwen3-1.7B base: sequential conditioning helps most where the model has enough knowledge to be corrected but not enough reliability to solve the problem in one independent attempt.

6 A Training-Time Probe

The inference-time result did not come from a successful multi-agent training run. As a secondary probe, we trained Qwen3-4B for 30 PPO updates with a role-asymmetric Solver-Verifier reward recipe. The resulting Solver reaches 0.634 single-shot accuracy on MATH-500, close to single-agent GRPO at 0.656. The full trained debate reaches only 0.574, far below the inference-time scaffolds on the untrained base. The Verifier also exhibits clear pathologies: among turns that emit a verdict tag, 96.5% are `approve`, and 53.4% of Verifier turns omit the tag entirely.

This is not evidence that training-time multi-agent methods fail in general. It is a bounded negative probe: this configuration did not outperform single-agent GRPO at this horizon, and the Verifier behavior suggests the intended training signal did not take hold. The result supports the narrower interpretation of the paper: the demonstrated effect is an inference-time sequential-conditioning effect, not a successful multi-agent training recipe. Full reward-design details are in Appendix E.

7 Discussion

The useful takeaway is not that debate should replace every single-agent baseline. It is that independent samples and conditioned revisions are different interventions. Independent sampling explores several first attempts from the same distribution. Sequential conditioning changes the context of the next attempt. On these Qwen checkpoints, that context change is often enough to reach answers absent from four independent samples. This is consistent with matched-thinking-budget critiques in which strong single-agent sequential reasoning can match or beat multi-agent systems: our central contrast is against independent parallel sampling, not against every possible single-agent reasoning scaffold.

Why does this setting differ from prior debate studies? Three differences are plausible. First, the scaffolds are strict and deterministic: four-turn alternations or one long chain-of-thought, rather than free-form agent conversations. Second, the comparison is against matched self-consistency, not an unmatched single-shot baseline. Third, the decomposition isolates the mechanism: long single-stream reasoning recovers most of the math lift, while explicit role-asymmetric debate adds a further 7–11 points on GPQA. The load-bearing primitive is sequential conditioning; critique-then-revise is a separable ingredient whose value depends on domain and checkpoint.

Why does base plus scaffold match solo-GRPO plus scaffold on math? The base and GRPO-trained models have different independent-sample distributions, but both appear to contain enough latent reasoning capacity for long-form sequential extraction to reach similar ceilings. The remaining difference is smaller and more specific: on the GRPO-trained checkpoint, Self-Refine and debate are about four points above long chain-of-thought, suggesting that outcome RL may make the model more responsive to explicit critique prompts.

The training-time probe sharpens the framing rather than undermining it. Our short multi-agent training run did not beat single-agent GRPO and showed clear Verifier pathologies. The positive result in this paper is therefore not a broad endorsement of multi-agent training. For these models and benchmarks, the productive effect is an inference-time scaffold, and the scaffold can often be reduced to a weaker commitment than debate: condition later reasoning on earlier reasoning.

8 Limitations

Scope. The positive finding is limited to Qwen-family thinking-mode-trained small models. The clean Llama-3.1-8B-Instruct probe in Appendix B is negative on both math and science, so model family is not a cosmetic caveat.

Domains. The non-math domain is GPQA-Diamond only. HumanEval, BBH, multi-hop QA, tool-use tasks, and interactive settings remain untested. Reflexion [12] and CRITIC [13] are cited but not run as full baselines.

Ablation coverage. Frozen-critic debate is run only on the GRPO-trained checkpoint. AIME cells have only 30 problems, so the intervals are wide even when the point estimates are large.

Sensitivity. We use $k = 4$ self-consistency with 2048 tokens per sample and fixed decoding settings. We do not sweep alternative parallel budget splits such as 8 shorter samples, adaptive-length sampling, temperatures, seeds, prompt variants, stronger multi-turn judges, or learned selectors over full rationales. These are direct follow-up baselines for testing how much of the sequential margin remains under a more optimized parallel or selector pipeline.

Evaluation artifacts. The original sympy-equivalence comparator short-circuited on unit-suffix mismatches such as 30° versus 30. We patched the comparator and rescored every transcript; residual bias from other unit suffixes is estimated at no more than 1 point absolute.

9 Conclusion

On Qwen-family thinking-mode-trained small models, sequential conditioning at matched generation budget substantially exceeds self-consistency: debate and Self-Refine gain +10 to +24 points on MATH-500 and +18 to +23 points on GPQA-Diamond, and debate exceeds the selector over four independent samples by +6 to +30 points on hard cells. The structural commitment that carries most of the value is not role asymmetry; it is long-form sequential reasoning instead of independent parallel sampling. Explicit critique and role-asymmetric debate add value in narrower regimes, especially science multiple-choice at base checkpoints. The Llama-3.1-8B probe loses to self-consistency, so the finding is scoped rather than universal. The natural follow-up is to test whether thinking-mode-style post-training is what makes a model receptive to these sequential scaffolds.

A Implementation Details

The 9-arm matrix is implemented in `dtca/eval/standalone.py`; arm dispatch via a `mode` field on `EvalConfig`. The asymmetric arm loads two co-located vLLM engines at half `gpu_memory_utilization` each. Aggregation helpers `majority_vote` and `any_correct` are unit-tested in `tests/test_eval/test_inference_arms.py` (107 tests, all passing). The comparator patch strips °, \circ, and \degree before sympy-equivalence fallback (`dtca/discussion/answer_parser.compare`). The DTCA training pipeline builds on Dr. MAS [21] / veRL; reward components live under `dtca/rewards/`, with each channel implementing a `RewardComponent` protocol that returns scalar value plus per-turn attribution for AT-GRPO-style gradient weighting.

B Cross-family probe — Llama-3.1-8B-Instruct

To test whether the matched-compute scaffold lift transfers off the Qwen family, we ran four arms (single-shot, debate, self-consistency, Self-Refine) on Llama-3.1-8B-Instruct — chosen as a deliberately non-thinking-mode-trained model of comparable scale (8B vs Qwen3's 4B), mid-2024 release, mature vLLM support, no documented post-training on rationalization/refinement trajectories. All numbers from the same eval pipeline plus post-rescore comparator.

Table 5: Llama-3.1-8B-Instruct cross-family probe ($n = 500$ for MATH-500, $n = 198$ for GPQA-Diamond).

Cell	Single	Debate	Self-Refine	Self-cons.	Debate − SC	SR − SC	SR − Debate
MATH-500	0.270	0.314	0.300	**0.348**	−3.4 [−7.4, +0.6]	−4.8 [−8.8, −0.6]	−1.4 null
GPQA-Diamond	0.126	0.182	0.172	**0.278**	−9.6 [−17.2, −2.0]	−10.6 [−18.2, −3.0]	−1.0 null

The matched-compute scaffold lift does *not* transfer to Llama-3.1-8B-Instruct. On both math and science, both critique scaffolds lose to self-consistency at matched compute, with three of four key CIs non-overlapping with zero. Self-Refine and debate remain statistically tied on math, but neither exceeds self-consistency; on GPQA both land near 0.17 against self-consistency's 0.278.

Two observations sharpen the story. First, single-shot accuracy is much lower than Qwen3-4B's (Llama 0.270 vs Qwen3-4B 0.544 on MATH-500); Llama is genuinely weaker at math at this scale, and absolute headroom for a scaffold is smaller. Second, the **L1 curse**: debate-self drops Llama from 0.744 single-shot L1 to 0.628 debate L1 — the scaffold *hurts* on easy problems where the model would have gotten the answer single-shot. Self-Refine does the same (L1: 0.744 $\rightarrow$ 0.581). On Qwen-family the L1 cells stayed at ceiling; on Llama the scaffold introduces noise that breaks them. We attribute this to Llama's instruction-tuning prior: when asked to "critique your prior answer", the model produces a critique even when the prior answer was correct, and the next turn re-conditions on a (sometimes spurious) critique.

The Qwen3-1.7B parallel is instructive: Qwen3-1.7B at 1.7B reaches 0.528 single-shot, while Llama-3.1-8B at 8B reaches only 0.270. Scale is not the explanatory variable; thinking-mode post-training is the candidate. Preliminary probes on Phi-3.5-mini-instruct (extraction-format mismatch), SmolLM2-1.7B-Instruct (8192-token context too small for 4-turn debate at 2048-per-turn), and Mistral-7B-Instruct-v0.3 (vLLM engine init bug) each hit a different infrastructure blocker; we report them only for completeness.

C Full per-level results

Table 6 reports per-cell accuracy on MATH-500 (per-level $n \in \{43, 90, 105, 128, 134\}$ for L1–L5) and AIME ($n = 30$ per year), post-rescore. Table 7 gives 95% paired bootstrap CIs on $p_B - p_D$, and Table 8 gives the same on $p_B - p_{\text{pass@4}}$.

Table 6: Per-cell accuracy: A single-shot, D self-consistency k=4, B 4-turn debate.

	A	D	B	B−A	B−D
Base Qwen3-4B					
Overall (n=500)	0.544	0.622	**0.858**	+31.4	+23.6
L1	0.930	0.907	0.930	+0.0	+2.3
L2	0.733	0.856	0.900	+16.7	+4.4
L3	0.648	0.724	0.895	+24.8	+17.1
L4	0.453	0.555	**0.859**	+40.6	+30.5
L5	0.299	0.358	**0.776**	+47.8	+41.8
AIME 2024	0.033	0.100	**0.367**	+33.3	+26.7
AIME 2025	0.067	0.000	**0.333**	+26.7	+33.3
Solo-GRPO step-30					
Overall (n=500)	0.668	0.752	**0.858**	+19.0	+10.6
L1	0.860	0.930	0.930	+7.0	+0.0
L2	0.778	0.867	0.878	+10.0	+1.1
L3	0.743	0.848	0.914	+17.1	+6.7
L4	0.633	0.719	**0.852**	+21.9	+13.3
L5	0.507	0.575	**0.784**	+27.6	+20.9
AIME 2024	0.133	0.100	**0.233**	+10.0	+13.3
AIME 2025	0.100	0.133	**0.367**	+26.7	+23.3
Base Qwen3-1.7B					
Overall (n=500)	0.528	0.572	**0.796**	+26.8	+22.4
L1	0.930	0.907	0.953	+2.3	+4.7
L2	0.689	0.778	0.878	+18.9	+10.0
L3	0.629	0.705	0.876	+24.8	+17.1
L4	0.477	0.492	**0.789**	+31.2	+29.7
L5	0.261	0.299	**0.634**	+37.3	+33.6
AIME 2024	0.000	0.033	**0.133**	+13.3	+10.0
AIME 2025	0.000	0.067	**0.267**	+26.7	+20.0

D Qualitative debate trajectory

The cleanest diagnostic of what debate adds beyond parallel-sample aggregation is the set of problems where the 4-turn debate produces the correct answer but all four pass@k samples are wrong. On solo-GRPO step-30 MATH-500 L5, 26 of 134 problems fall in this category. One representative trajectory follows.

The problem asks: "The expression $2 \cdot 3 \cdot 4 \cdot 5 + 1$ is equal to 121. However, we can obtain values other than 121 by inserting parentheses (e.g. $(2 \cdot (3 \cdot 4)) \cdot (5 + 1) = 144$). In total, how many values can be obtained?" Ground truth: 4.

Table 7: Per-cell 95% paired bootstrap CIs on $p_B - p_D$ (2000 resamples), post-rescore.

Cell	$p_B - p_D$	95% CI
Base Qwen3-4B MATH-500 overall	+23.6	[+19.4, +27.8]
L4	+30.5	[+22.7, +39.1]
L5	+41.8	[+33.6, +50.7]
AIME 2024	+26.7	[+13.3, +43.3]
AIME 2025	+33.3	[+16.7, +50.0]
Solo-GRPO MATH-500 overall	+10.6	[+7.4, +14.0]
L4	+13.3	[+5.5, +21.1]
L5	+20.9	[+14.2, +28.4]
AIME 2024	+13.3	[+3.3, +26.7]
AIME 2025	+23.3	[+10.0, +40.0]
Qwen3-1.7B MATH-500 overall	+22.4	[+18.4, +26.4]
L4	+29.7	[+21.1, +38.3]
L5	+33.6	[+25.4, +41.8]
AIME 2024	+10.0	[+0.0, +20.0]
AIME 2025	+20.0	[+6.7, +33.3]

Table 8: Per-cell 95% paired bootstrap CIs on $p_B - p_{\text{pass@4}}$ (2000 resamples), post-rescore.

Cell	$p_B - p_{\text{pass@4}}$	95% CI
Base Qwen3-4B MATH-500 overall	+15.2	[+11.4, +19.0]
L5	+29.9	[+21.6, +38.8]
AIME 2025	+20.0	[+6.7, +36.7]
GPQA-Diamond	+17.2	[+9.6, +24.7]
Solo-GRPO MATH-500 overall	+6.0	[+3.0, +9.2]
L5	+17.9	[+11.2, +25.4]
AIME 2025	+23.3	[+10.0, +40.0]
GPQA-Diamond	+12.1	[+3.5, +19.2]
Qwen3-1.7B MATH-500 overall	+10.8	[+7.6, +14.4]
L5	+19.4	[+11.2, +27.6]
AIME 2025	+16.7	[+3.3, +30.0]
GPQA-Diamond	+15.2	[+7.6, +22.7]

The four parallel pass@4 samples gave boxed answers of 60, 122, 126, and 144; none correct. Each found a different way to misread the problem, counting expression evaluations rather than distinct numerical values. The debate trajectory unfolded differently. At T0 the Solver boxed 14, having reasoned through some parenthesizations but lost track of the "distinct values" framing. At T1 the Verifier critiqued by enumerating specific parenthesizations such as $(2 \cdot 3 \cdot 4) \cdot (5 + 1) = 144$ and $2 \cdot (3 \cdot (4 \cdot 5 + 1)) = 126$; its own boxed answer of 126 was wrong, but its critique named the operation as enumerating groupings and reading off distinct numerical values. At T2 the Solver re-attempted with the Verifier's framing, listed ten parenthesizations, collected the distinct values $\{121, 122, 126, 144\}$, and boxed the correct answer 4. At T3 the Verifier approved 4.

The Verifier's T1 critique did not contain the correct answer. What it contained was a framing of the task (enumerate groupings, count distinct values) that was the missing piece in the Solver's T0 reasoning. This is the mechanism Section 5.3 names: the critique turn supplies reasoning structure the Solver did not surface single-shot, and the next Solver turn re-conditions on that structure.

E DTCA Training-Time Probe Details

The DTCA probe uses a role-asymmetric four-turn Solver-Verifier protocol. Verifier turns are asked to emit both a verdict and a boxed answer. The reward design separates verdict quality from final-answer correctness, includes a counterfactual Verifier-value channel, normalizes channels with GDPO-style per-channel advantages [21], and applies a raw format penalty when the Verifier omits required tags.

The two recipes produce Solvers closely matched on single-shot accuracy at 30 PPO updates (DTCA 0.634, solo-GRPO 0.656 on MATH-500 overall; per-level deltas above). The principal DTCA Verifier pathologies are characterized quantitatively in Section 6: a 96.5% rubber-stamp approval rate on the

Table 9: DTCA vs solo-GRPO per-level lift over base on MATH-500.

Level	Base	Solo-GRPO Δ	DTCA Δ
L1	0.907	−0.070	−0.008
L2	0.722	+0.045	+0.030
L3	0.629	+0.104	+0.124
L4	0.445	+0.172	+0.203
L5	0.299	+0.201	+0.147

Verifier turns that emit a `<verify>` tag, and a 53.4% missing-tag rate overall. The raw-bypass per-turn format penalty addresses the second pathology by adding a fixed -1.0 reward to per-turn advantages bypassing GDPO per-channel normalization, restoring gradient pressure on format compliance once the in-channel penalty loses traction. The approval-bias pathology is more fundamental: with Solver correctness at base rate ≈ 0.65 on MATH L3–5, always-approve has unconditioned expected reward $\approx +0.30$ before class reweighting and remains positive after the EMA-based Self-Verify weights settle. A principled fix is left to future work; the pre-RL Solver-T0 lift in Table 9 is therefore not attributable to a working Verifier — the Verifier's training signal is largely a slow constant that the GRPO update absorbs through the shared-policy Solver gradient.

References

[1] Hangfan Zhang, Zhiyao Cui, Jianhao Chen, Xinrun Wang, Qiaosheng Zhang, Zhen Wang, Dinghao Wu, and Shuyue Hu. Stop overvaluing multi-agent debate – we must rethink evaluation and embrace model heterogeneity. *arXiv preprint arXiv:2502.08788*, 2025.

[2] Jiawei Xu, Arief Koesdwiady, Sisong Bei, Yan Han, Baixiang Huang, Dakuo Wang, Yutong Chen, Zheshen Wang, Peihao Wang, Pan Li, and Ying Ding. Rethinking the value of multi-agent workflow: A strong single agent baseline. *arXiv preprint arXiv:2601.12307*, 2026.

[3] Dat Tran and Douwe Kiela. Single-agent LLMs outperform multi-agent systems on multi-hop reasoning under equal thinking token budgets. *arXiv preprint arXiv:2604.02460*, 2026.

[4] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[5] Chanwoo Park, Seungju Han, Xingzhi Guo, Asuman Ozdaglar, Kaiqing Zhang, and Joo-Kyung Kim. MAPoRL: Multi-agent post-co-training for collaborative large language models with reinforcement learning. *arXiv preprint arXiv:2502.18439*, 2025.

[6] Shuo Liu, Tianle Chen, Zeyu Liang, Xueguang Lyu, and Christopher Amato. LLM collaboration with multi-agent reinforcement learning. *arXiv preprint arXiv:2508.04652*, 2025. Introduces MAGRPO (Multi-Agent Group Relative Policy Optimization).

[7] Yujie Zhao, Lanxiang Hu, Yang Wang, Minmin Hou, Hao Zhang, Ke Ding, and Jishen Zhao. Stronger-MAS: Multi-agent reinforcement learning for collaborative LLMs. *arXiv preprint arXiv:2510.11062*, 2025. Introduces the AT-GRPO (Agent- and Turn-wise grouped) algorithm.

[8] Kaiyan Zhang, Kai Tian, Runze Liu, Sihang Zeng, Xuekai Zhu, Guoli Jia, Yuchen Fan, Xingtai Lv, Yuxin Zuo, Che Jiang, Yuru Wang, Jianyu Wang, Ermo Hua, Xinwei Long, Junqi Gao, Youbang Sun, Zhiyuan Ma, Ganqu Cui, Ning Ding, and Biqing Qi. MARTI: A framework for multi-agent LLM systems reinforced training and inference. In *ICLR*, 2026.

[9] Chenxi Whitehouse, Tianlu Wang, Ping Yu, Xian Li, Jason Weston, Ilia Kulikov, and Swarnadeep Saha. J1: Incentivizing thinking in LLM-as-a-judge via reinforcement learning. *arXiv preprint arXiv:2505.10320*, 2025.

[10] Fuxiang Zhang, Jiacheng Xu, Chaojie Wang, Ce Cui, Yang Liu, and Bo An. Incentivizing LLMs to self-verify their answers. *arXiv preprint arXiv:2506.01369*, 2025.

[11] Xiaoying Zhang, Yipeng Zhang, Hao Sun, Kaituo Feng, Chaochao Lu, Chao Yang, and Helen Meng. Critique-GRPO: Advancing LLM reasoning with natural language and numerical feedback. *arXiv preprint arXiv:2506.03106*, 2025.

[12] Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[13] Zhibin Gou, Zhihong Shao, Yeyun Gong, Yelong Shen, Yujiu Yang, Nan Duan, and Weizhu Chen. CRITIC: Large language models can self-correct with tool-interactive critiquing. In *International Conference on Learning Representations (ICLR)*, 2024.

[14] John Seon Keun Yi, Aaron Mueller, and Dokyun Lee. Latent agents: A post-training procedure for internalized multi-agent debate. *arXiv preprint arXiv:2604.24881*, 2026.

[15] Yinyi Luo, Yiqiao Jin, Weichen Yu, Mengqi Zhang, Srijan Kumar, Xiaoxiao Li, Weijie Xu, Xin Chen, and Jindong Wang. AgentArk: Distilling multi-agent intelligence into a single LLM agent. *arXiv preprint arXiv:2602.03955*, 2026.

[16] Weizhen Li, Jianbo Lin, Zhuosong Jiang, Jingyi Cao, Xinpeng Liu, Jiayu Zhang, Zhenqiang Huang, Qianben Chen, Weichen Sun, Qiexiang Wang, Hongxuan Lu, Tianrui Qin, Chenghao Zhu, Yi Yao, Shuying Fan, Xiaowan Li, Tiannan Wang, Pai Liu, King Zhu, He Zhu, Dingfeng Shi, Piaohong Wang, Yeyi Guan, Xiangru Tang, Minghao Liu, Yuchen Eleanor Jiang, Jian Yang, Jiaheng Liu, Ge Zhang, and Wangchunshu Zhou. Chain-of-agents: End-to-end agent foundation models via multi-agent distillation and agentic RL. *arXiv preprint arXiv:2508.13167*, 2025.

[17] Pei Yang, Ke Zhang, Ji Wang, Xiao Chen, Yuxin Tang, Eric Yang, Lynn Ai, and Bill Shi. Multi-agent collaborative reward design for enhancing reasoning in reinforcement learning. *arXiv preprint arXiv:2511.16202*, 2025. Introduces CRM (Multi-Agent Collaborative Reward Model).

[18] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

[19] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? *arXiv preprint arXiv:2504.13837*, 2025.

[20] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. Introduces Group Relative Policy Optimization (GRPO).

[21] Lang Feng, Longtao Zheng, Shuo He, Fuxiang Zhang, and Bo An. Dr. MAS: Stable reinforcement learning for multi-agent LLM systems. `https://github.com/feng-y16/Dr.MAS`, 2026. Code release; companion preprint forthcoming.